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Syn-extension rotations in the Permian St Affrique basin (Massif Central, France): paleomagnetic constraints

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ABSTRACT

We present the results of a paleomagnetic study of the Permian sedimentary basin of St Affrique (southern Massif Central, France). This study was devised to test for tectonic rotations in the Montagne Noire area during the Stephano-Permian extension in the Hecynian chain and to complete a preliminary study [1] in which we suggested an counterclockwise rotation of St Affrique basin with respect to the neighbouring Lodève basin. This rotation was attributed to the domino-like rotation of the basement of St Affrique (i.e., the Paleozoic Monts de Lacaune imbricate units) as expected from the tectonic interpretation of the Montagne Noire area made by Van Den Driessche and Brun [2,3]. However, the rotation of the basin during Permian should induce declination differences between the basal and the upper part of the stratigraphic sequence. We therefore conducted vertical sampling of the basin throughout the Permian sequence. Two sites of Stephanian age gave no results, but twelve sites from the lower to the upper Permian have been sampled in the St Affrique basin and did provide results. Paleomagnetic analysis using thermal procedures confirms our previous analysis and allowed us to demonstrate three magnetization components. The low temperature A component is unblocked by 300°C and conforms to the present-day dipole field direction. An intermediate temperature component (B) has also been isolated, with unblocking temperatures of 300° to 550°–600°C, and this component was possibly carried by magnetite. Between 600° and 680°C a third component (C) may be suspected at some sites, but it has not been isolated with confidence. At a site showing a tilted-block geometry produced by syn-sedimentary normal faulting (site p7), it is shown that the B component is an early magnetization acquired during faulting of the succession, and we are therefore able to give it a stratigraphic age. The mean directions of the B component, which are computed at the stratigraphic stage level, show the following counterclockwise deviations from the reference magnetic field direction for stable Europe: $21° \pm 4.5°$ for the upper Autunian, $10° \pm 7°$ for the Saxonian, $10° \pm 9°$ for the lower Thuringian and $4° \pm 6°$ and $3.5° \pm 4.5°$ for two successive stages of middle/upper Thuringian age. We thus consider that (1) the comparison of the lowest units of St Affrique and Lodève basins [1], and (2) the evolution in paleomagnetic declination during the Permian within the St Affrique basin itself provide evidence, in space and time, of a rotation of the basin during its formation. We attribute this rotation to the domino-like rotation of its basement (the Monts de Lacaune imbricate units) during the Stephano-Permian extension in the Hercynian chain [2,3].

1. Introduction

This study was undertaken in order to complete a preliminary study on the paleomagnetism of the Permian St Affrique sedimentary basin (Massif Central, France) [1]. Our goal in this study, is to test the hypothesis of a rotation about a vertical axis of the basin during its formation. This hypothesis has been inferred from the tectonic evolution of the Montagne Noire region during the Stephano-Permian extension in the Variscan mountain belt, as proposed by Van Den Driessche and Brun [2,3].

The extensional system of the Montagne Noire developed during Stephano-Permian post-thickening extension. It consists of a north dipping major detachment, named the Espinouse fault, which is located along the northern border of the Montagne Noire gneiss dome (Fig. 1a). The hangingwall of this extensional system is formed by the Paleozoic metasedimentary rocks of the Monts de Lacaune and Albigeois nappes, which constitute the basement of the St Affrique basin. Isostatic rebound during extension induced the uplift of ductile lower crust in the footwall, resulting in the formation of the Montagne Noire

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migmatitic gneiss dome. The Espinouse detachment is connected with flat-lying shearing in the lower crust resulting in a large scale roll-over curvature of the hangingwall. This shearing deformation was responsible (1) for the reactivation as low angle, listric normal faults of the previous

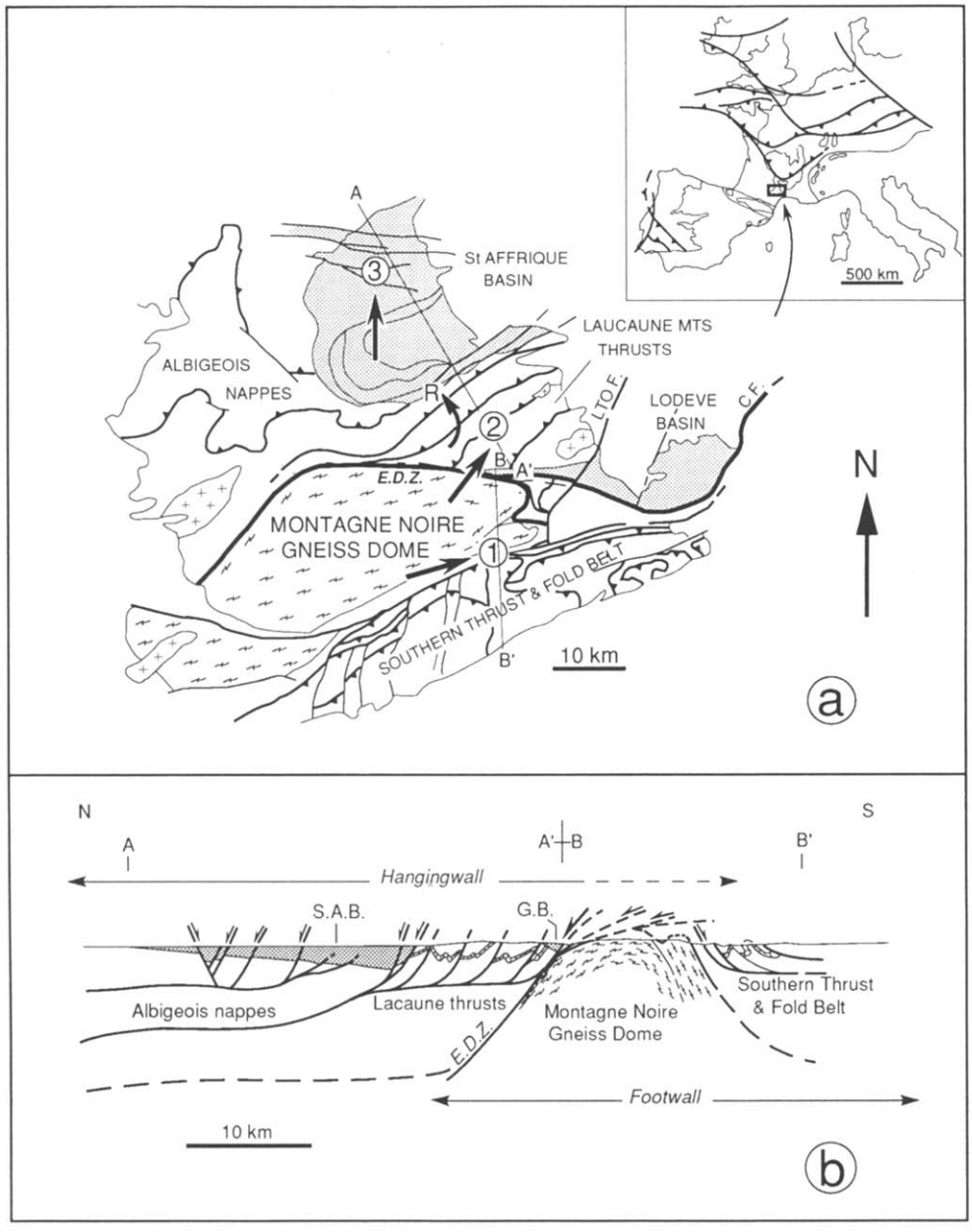


Fig. 1. (a) Tectonic map of the studied area. *E.D.Z.* = Espinouse detachment zone; *LTO F.* = La Tour-sur-Orb fault; *C F.* = Cévennes fault. The black arrows show the direction of extension during Namurian (or older), Stephanian and mid- to Late Permian times (number 1, 2 and 3 respectively) and emphasize the progressive migration of extension toward the north. The curved black arrow (*R*) indicates the suspected rotation of the Monts de Lacaune basement rocks. (b) Schematic cross section through the studied area showing the location of the Permo-Carboniferous basins in the Montagne Noire extensional system. *S.A.B.* = St Affrique basin; *G.B.* = Graissessac (western Lodève) basin.

Varican thrusts of the hangingwall, and (2) for the strong asymmetry of the Stephano-Permian basins, which is emphasized by the pervasive southward tilting of their infill (Fig. 1b).

Kinematic analysis of ductile deformation and radiometric data in the footwall show that the direction of extension varied from ESE to N–S from Stephanian to Permian times. In the hangingwall the steeply dipping (to the northwest) thrusts of the Monts de Lacaune are classically related to the main Variscan phase of crustal thickening [4–7]. However, their NE–SW orientation is anomalous with regards to the general E–W trend of thrust faults in the rest of the Massif Central [4]. This led Van Den Driessche and Brun [2,3] to suggest that the present orientation of the Monts de Lacaune thrusts results from a counterclockwise, domino-like rotation induced

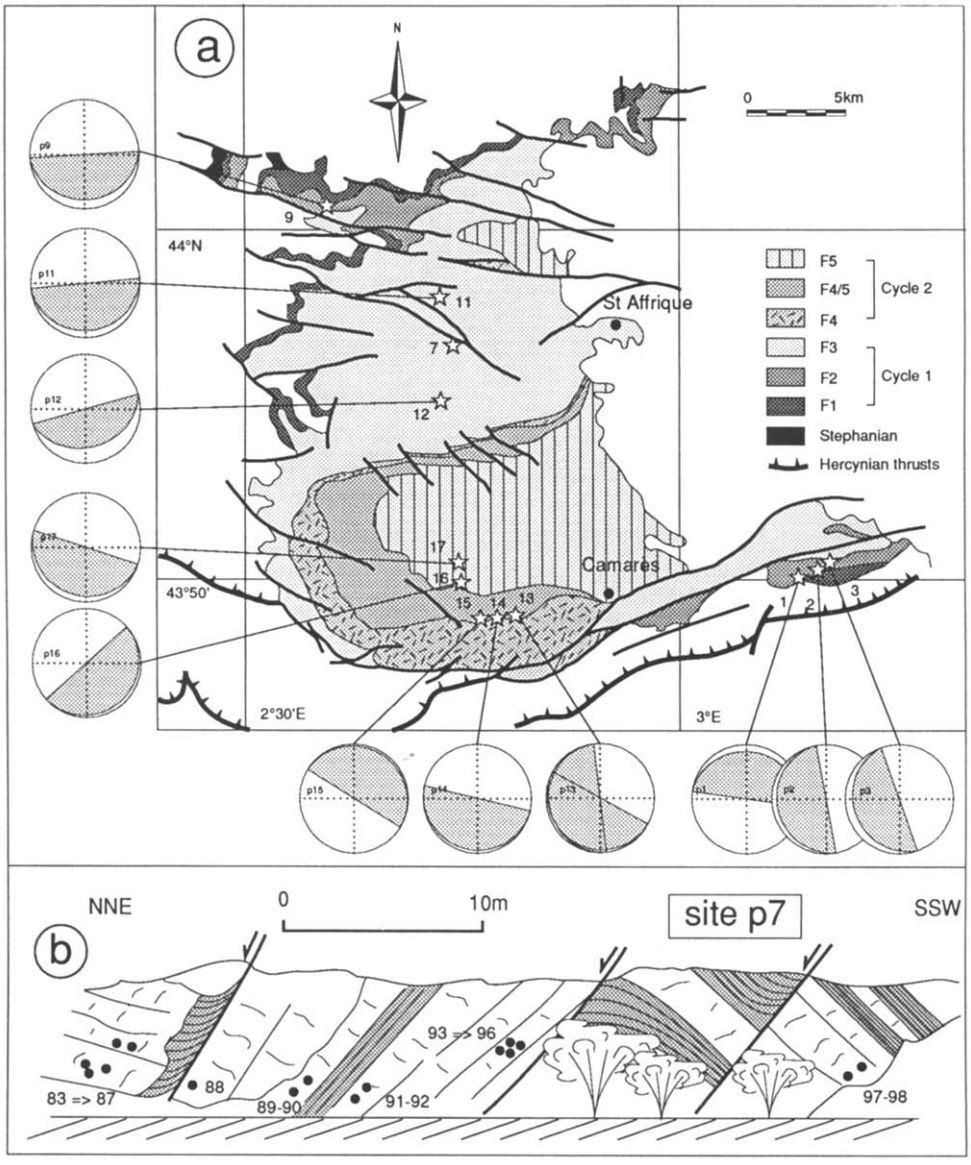


Fig. 2. (a) Schematic geological map of the St Affrique basin (after Legrand [8]). Boxes F1 to F5 show the five successive formations described by Legrand [8]; white numbered stars are the sampling sites; stereonets show bedding attitude at each site. (b) Schematic view of site p7 outcrop, with location of cores 83 to 98, showing the normal fault structure interpreted by Legrand [8] as due to extension on the outer edge of a roll-over anticline.

by the progressive change in the direction of extension, from ENE–WSW to N–S, during late Carboniferous to Permian times (arrows 1,2 and 3 in Fig. 1a). We therefore attempt to test this rotation of the Monts de Lacaune imbricate units by using paleomagnetic data.

Strong ductile deformation and low grade metamorphism in the Cambro-Silurian units of the Monts de Lacaune do not allow us to perform conclusive paleomagnetic studies. Fortunately, extension is accompanied by the development of Stephano-Permian sedimentary basins throughout the Hercynian chain. In our region of interest, we focussed our attention on two of these basins, the St Affrique and the Lodève. As mentioned above, the structural pattern of the St Affrique basin is controlled by low angle normal faulting in the basement, which corresponds to the reactivation of Variscan thrusts of both the Albigeois nappes and Monts de Lacaune units. Assuming that the latter underwent counterclockwise rotation during extension, this implies that the syn-tectonic deposits of the St Affrique basin must record such progressive rotation. In contrast, the structural pattern of the Lodève basin is directly controlled by the Espinouse detachment and the NE–SW directed La Tour-sur-Orb and Cévennes faults, which act as transfer faults during extension (Fig. 1a). In this area no rotation was expected, as demonstrated by the E–W direction of thrust related fabrics in the basement.

Results obtained in the preliminary paleomagnetic study of the Autunian formations [1] showed about 20° of counterclockwise declination difference in the base of the St Affrique basin with respect to the Lodève basin; this fits the reference field direction for stable Europe. We interpreted this difference as a counterclockwise rotation of St Affrique with respect to Lodève during the Permian.

However, apart from a declination difference between the base of the two basins, a second important consequence should arise from the Monts de Lacaune rotation. In effect, the rotation observed at the base of the St Affrique basin should diminish as we proceed up the succession, reaching zero at the top of the Permian sedimentary column. We therefore completed the preliminary study by vertical sampling of the whole Permian succession of the St Affrique basin. The results of the paleomagnetic analysis of these samples are given below.

2. Geology and sampling sites

According to Legrand [8] and Rolando et al. [9], the whole Permian series of the St Affrique basin may be divided into five distinct continental formations, which in turn may be grouped into two main cycles (Fig. 2a).

The first cycle comprises three successive formations, F1 to F3. The F1 formation, which is made up of conglomerates, sandstones and pelites, shows an alluvial fan facies. The F2 formation is divided into a dark pelitic lacustrine facies part (F2a), and a grey and red, pelitic and

TABLE 1

Chronostratigraphic and structural data of St Affrique sites

Formation	Stage [a]	Age [b] (Ma)	Site	Bedding (strike/dip)	Site location	
F1–F2a	upper Autunian	280–272	p1	278/14	43°50′35″N	03°01′01″E
			p2	168/10	43°51′18″N	03°03′18″E
			p3	160/16	43°51′20″N	03°03′20″E
F2b	Saxonian	272–258	p9	86/18	44°00′02″N	02°38′54″E
F3	lower Thuringian	258–254	p7	see Table 3	43°57′00″N	02°45′28″E
			p11	84/14	43°57′42″N	02°44′54″E
			p12	74/30	43°55′00″N	02°44′40″E
F4/5	middle/upper Thuringian	254–245	p13	170/6–119/7	43°48′08″N	02°49′30″E
			p14	103/6	43°48′22″N	02°48′44″E
			p15	302/5	43°48′32″N	02°47′31″E
F5	middle/upper Thuringian	254–245	p16	48/7	43°49′53″N	02°44′58″E
			p17	107/12	43°50′23″N	02°44′49″E

[a] After Legrand [8]. [b] Age boundaries between Autunian, Saxonian and Thuringian after Odin and Odin [22].

fluviatile sandstone part (F2b). The uppermost formation of the first cycle, F3, shows a fluviatile facies, and is made up of red pelites and sandstones. Dating of these formations is not easy, due to the oxidizing conditions of deposition. However, a chronostratigraphy, based on palynological analysis, has been proposed by Rolando et al. [9]. The F1 and F2a formations have been proposed as having an Autunian age. The F2b formation is not dated. However, due to its stratigraphic position it is attributed a Saxonian age. F3 is dated to the lower Thuringian.

The second cycle, which is separated from the first by an erosion surface and an angular unconformity, begins with the F4 formation, an alluvial fan facies of red conglomerates and sandstones. The F5 formation, the last of the St Affrique basin, is made up of red pelites with a few sandstone beds. Neither of these formations is dated. However, they succeed to F3, and are unconformably overlain by Triassic sediments to the east. They are therefore believed to be of middle to upper Thuringian age.

Because of our aim of following paleomagnetic

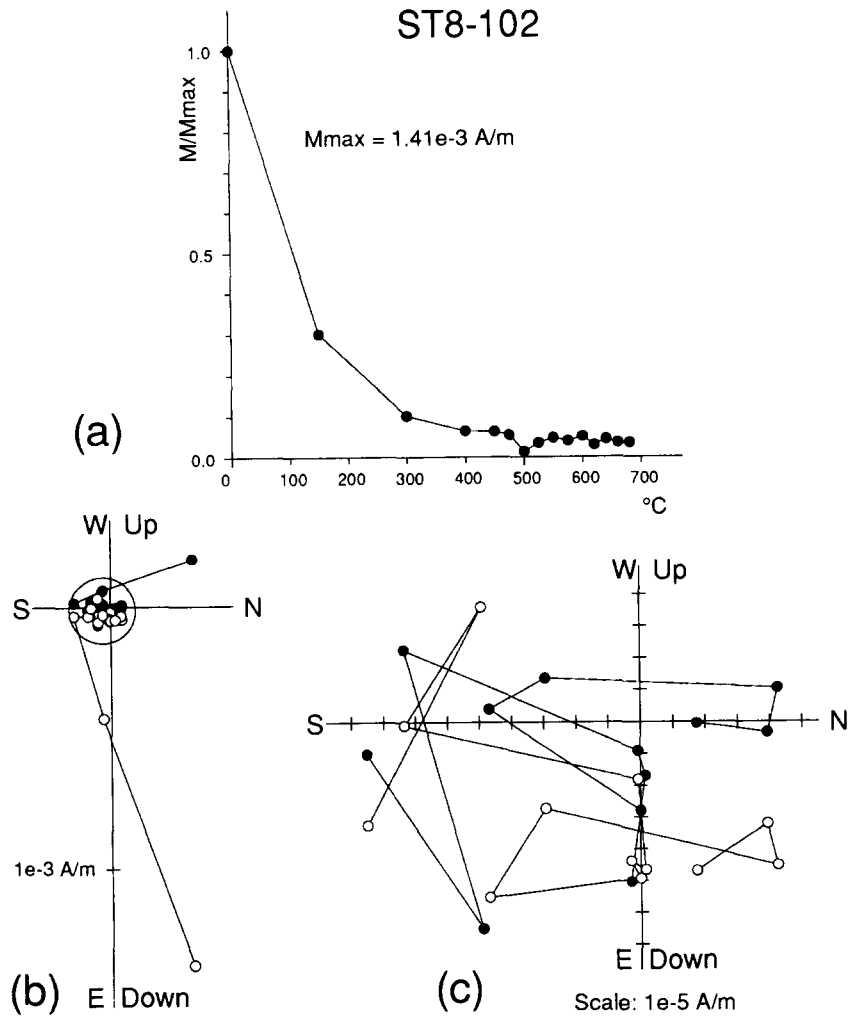


Fig. 3. Typical example of thermal demagnetization curves for Stephanian specimens. (a) Normalized magnetization intensity decay curve. (b) Orthogonal vector end-point projection [25] for *in-situ* data. (c) Enlargement of the area encircled in (b), for temperatures greater than 300°C. Closed symbols = projection onto the horizontal plane; open symbols = projection onto the vertical plane. Temperatures in °C.

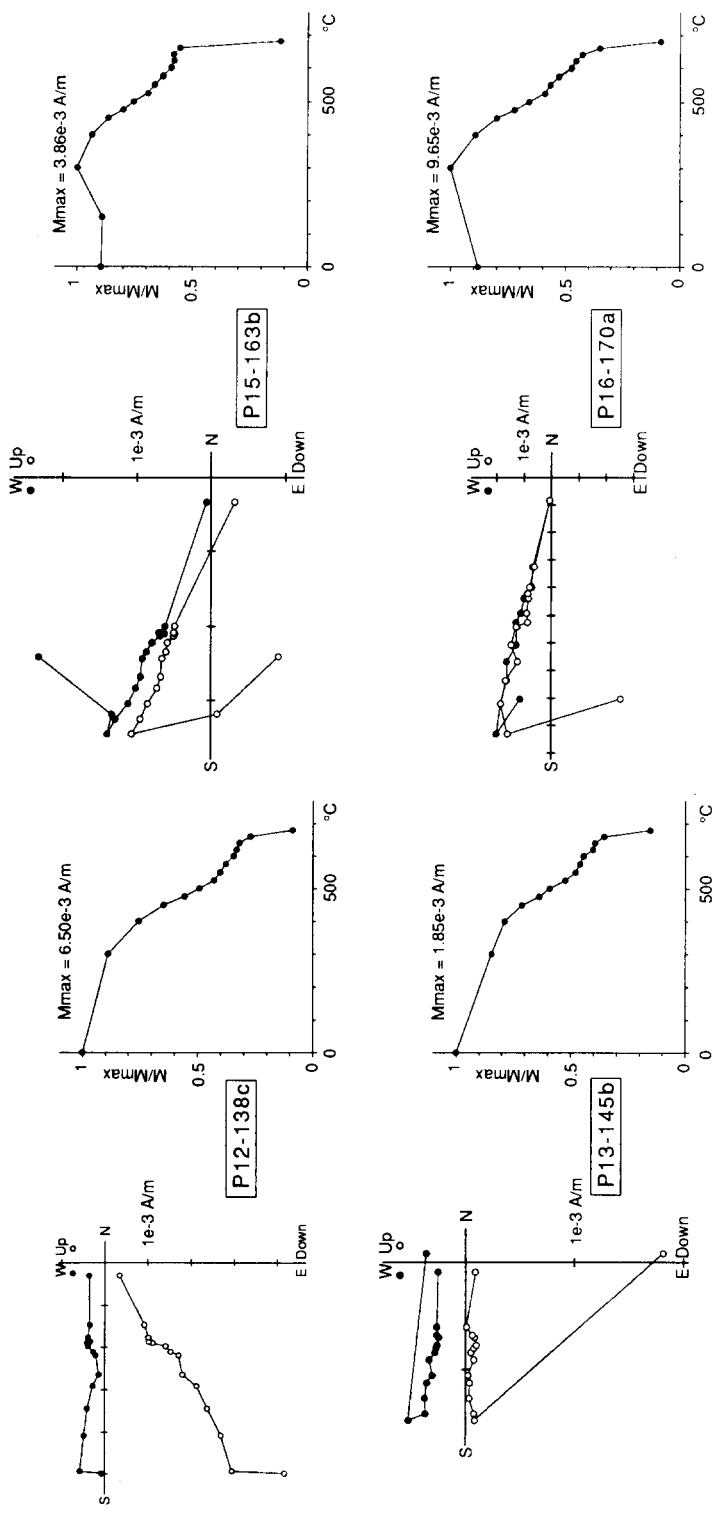


Fig. 4. Typical examples of thermal demagnetization curves for one specimen from each Permian site. *Left* Orthogonal vector end-point projection [25] for *in-situ* data. *Right* Normalized intensity decay curve. Same conventions as in Fig. 3.

vector evolution during the Permian, a total of 120 oriented cores have been drilled at twelve sites (including sites 1, 2 and 3 of our preliminary study [1]) distributed over the five Permian formations (Table 1 and Fig. 2a). Sites p1, p2 and p3 are grey upper Autunian sandstones, site p9 was drilled in grey-red Saxonian sandstones. All other sites of Permian age, p7 and p11 to p17, come from typical red purple sandstones of Thuringian age. Finally, two additional sites, st8 and st10, have been sampled in the grey Stephanian (upper Carboniferous) microconglomerates that outcrop in the northwestern part of the basin.

Local post-Permian (Alpine) deformation may be observed, but compared to syn-sedimentary structures is of minor importance [8]. It is characterized by normal faults at the northern border of the basin, and metre-size folds due to layer buckling that are located in the F3 and F5 formations at the southern border of the basin. Generally speaking, the basin is characterized by an onlap

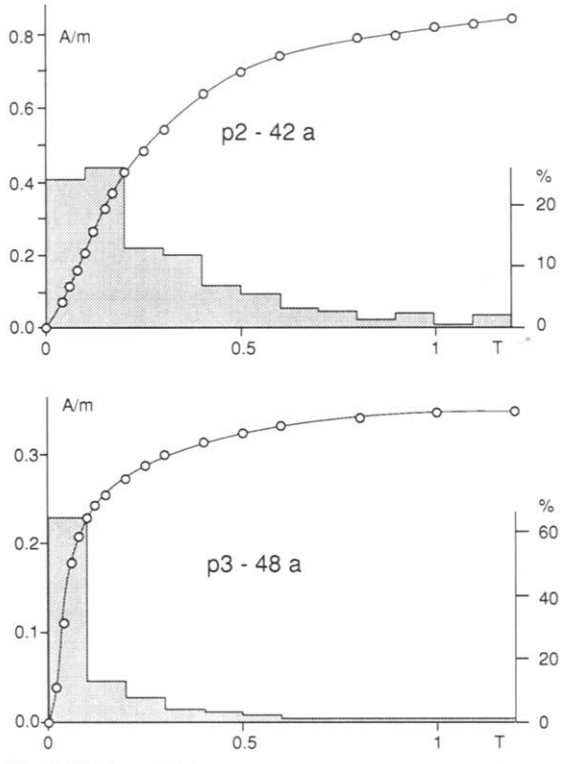


Fig. 5. IRM acquisition curves and coercivity spectra (in grey) for one sample each from sites p2 and p3. Field in Tesla; left scale—IRM intensity in A/m, right scale—percentage of IRM at 1.2 T.

structure [2] induced by the progressive deepening of the basin during extension; the beds dip slightly to the south (Table 1, Fig. 1b). These features produce internal angular unconformities between different formations and also within each individual formation.

Site p7 deserves special attention. This site was sampled along a 50 m long cross section where normal faults showing displacements of about a metre produced a tilted-block geometry, with the bedding within the blocks dipping either to the south or to the north. The cross section of this outcrop is shown in Fig. 2b. This peculiar structure has been interpreted [8,10,11] as a crestal collapse of blocks at the crest of a roll-over anticline structure due to basin extension on a flat listric major fault, following the models of McClay and Ellis [12]. This structure was formed by a very early event, simultaneously with bed deposition. A sampling of beds showing dips of various attitudes should therefore allow us to test for the age of magnetization, in a way similar to the classical fold test. Sixteen cores have been drilled at site p7.

3. Paleomagnetic analysis

3.1 Demagnetizations and IRM

A total of 129 specimens, including those of the preliminary study [1], have been thermally demagnetized, and measured using a CTF cryogenic magnetometer.

Ten pilot demagnetizations have been conducted on specimens from Stephanian sites st8 and st10. A typical example of the behaviour of magnetization in these specimens is shown in Fig. 3. More than 90% of the natural remanent magnetization (NRM) is lost by 300°C (Fig. 3a). The isolated low temperature component (below 300°C, Fig. 3b), has a northerly declination, and a steep downward inclination, very close to the present-day magnetic dipole field direction. It is thus considered as a recent overprint. Although removal of this overprint gives the magnetization a southerly direction with low inclination (Fig. 3b), consistent with the Upper Carboniferous/Lower Permian pole of Europe [13], the signal becomes very noisy (Fig. 3c), and it has not been possible to determine the direction of the high

temperature magnetization component with good confidence. All the results from Stephanian sites have therefore been rejected for any further analysis.

Typical demagnetization curves of specimens from the Permian formations F1 to F5 are shown in Fig. 4. The analysis of these curves confirms the previous one [1]. Apart from a recent overprint (A component) that contributes a few percent to the total NRM, and which is unblocked by 300°C (e.g. Fig. 4, specimens p13-145b, p14-152), it appears that NRM is systematically carried by two families of magnetic carriers with distinct blocking temperature spectra; this is particularly obvious in the normalized intensity decay curves. The first carrier progressively demagnetizes below 600°C, and represents more than 50% of magnetization intensity. We can then observe a stability in magnetization intensity, with no change in direction (e.g. Fig. 4, specimens p12-138c, p15-163b, p17-174c), followed by a sharp breakdown at about 670–680°C.

IRM acquisition curves and coercivity spectra (Fig. 5) also show the presence of two magnetic minerals. A low coercivity one is saturated by a field of 0.1 T, and may contribute to 60–70% of the total IRM at 1.2 T (Fig. 5, p3-48a), but there is a continuing acquisition by a high coercivity phase up to fields of 1.2 T. The relative proportion of both minerals is variable, as may be seen from the comparison of IRM curves of Fig. 5.

The results of the thermal demagnetization and IRM experiments are consistent with the occurence of two significant magnetic phases within the St Affrique samples. We attribute the high unblocking temperatures and high coercivities to hematite, which is indeed present in the redbeds specimens, but which also occurs in the grey specimens of sites p1, p2, p3 and p9. Magnetite is the likely carrier, with intermediate unblocking temperatures (300° to 550–600°C) and low coercivities.

3.2 Magnetization components

Examination of the orthogonal projections of Fig. 4 clearly shows that both intermediate and high unblocking temperature magnetic minerals carry a NRM component with southerly directions and shallow inclinations. However, apart from a few examples (Fig. 4; p14-152, p16-170a), demagnetization points between 300 and 600°C rarely converge exactly towards the origin. A medium temperature component, named the B component in the following, has therefore been determined by fitting lines on the linear parts of the demagnetization diagrams between 300–400°C and 550–600°C. Mean site directions of this characteristic remanent magnetization B component are given in Table 2.

Above 600°C, magnetization remains stable in intensity and direction and then sharply de-

TABLE 2

Site mean directions of the B component *

Site	n/N	*In-situ*			Tilt-corrected		
		Dec / Inc	k	α_{95}	*Dec / Inc*	k	α_{95}
p1	11/14	173.5/8.0	46	7.0	172.5/21.0	46	7.0
p2	9/11	172.0/11.5	91	5.4	174.0/11.5	93	5.4
p3	9/11	173.0/−4.0	138	4.4	171.5/−7.5	138	4.4
p7	15/16	189.0/−8.0	9	13.7	188.5/4.0	13	11.1
p9	8/9	185.0/−3.0	118	5.1	186.0/−20.5	118	5.1
p11	8/8	185.0/12.0	23	11.8	185.0/−1.5	23	11.8
p12	6/8	184.5/27.0	148	5.5	182.0/−1.5	148	5.5
p13	7/8	193.5/−5.5	90	6.4	193.0/−9.5	100	6.1
p14	7/8	199.0/−5.5	224	4.0	199.0/−11.5	224	4.0
p15	7/8	204.5/−3.0	20	13.7	204.5/2.0	20	13.7
p16	9/9	190.5/−9.0	126	4.6	191.5/−13.5	126	4.6
p17	9/9	192.5/−3.5	156	4.1	192.0/−15.5	156	4.1

* n/N = number of entries in the statistics/number of studied specimens; *dec / inc* = declination/inclination of mean site directions of magnetization; k and α_{95} = Fisher [17] statistics parameters.

creases at about 670–680°C. Owing to the fact that the B component does not always converge towards the origin, we suspect the presence of a third, high temperature component (C) carried by hematite. However, because successive demagnetization steps between 600° and 680°C remove no remanence, we cannot assume that a single component has been isolated, and this third, high temperature, C component will not therefore be considered in the following.

4. Analysis of paleomagnetic directions

4.1 Site p7: age of magnetization

Owing to the lack of folds, and very slight dipping of the beds, dating of magnetization is one of the main problems encountered in the Permian Lodève and St Affrique basins [1,14,15]. Apart from localized post-Permian deformation, the dip is generally due to the progressive deepening of the basin during its opening on a normal listric fault. This gives the basin an onlap structure [2,8] with slight internal unconformities between beds. However, one of the most striking structural features in basins such as these is the possible development of hangingwall roll-over anticlines [e.g., 16,12], where the extension at the hinge of the anticline is accommodated by the development of normal faults with synthetic and antithetic block rotations. In the St Affrique basin, such a crestal collapse of blocks structure may be found near the Le Dourdou du Viala locality, on the D133 road [8]. We have sampled sixteen oriented cores in this structure (Fig. 2b), which is one of the few places where we find beds dipping significantly to both the north and the south at the outcrop scale. (It is important to emphasize that a syn-sedimentary structure such as that described here develops very quickly after deposition of the sediments.)

Directions of the B component determined in each specimen from site p7 are given in Table 3, together with bedding plane directions. From these data, it is clear that there exists a relationship between the dip of the beds and magnetization inclination: northerly dipping beds have negative magnetization inclination (specimens 88 to 96), while southerly dipping beds (83 to 87 and 97 and 98) show positive inclinations. However, the complete tilt correction does not induce a significant clustering of directions at the outcrop scale (Tables 2 and 3). We therefore attempted a stepwise untilting of these directions, in a manner similar to the well-known stepwise unfolding. The results of this stepwise untilting, using 5% steps, are illustrated in Fig. 6, where the grouping parameter k is expressed as a function of untilting percentage. Because there is a dispersion in declination, which is also related to the groups of specimens, we present the results of both Fisher [17] statistics on directions (Fig. 6, top) and McFadden and Reid [18] statistics on inclinations (Fig. 6, bottom). From Fig. 6, it is clear that the vector population clusters at about 60% of untilting. In Fisher [17] statistics this clustering is significant at the 99% confidence level [19] with respect to the uncorrected distributions, and at the 90% confidence level with respect to the fully tilt-corrected directions. The evolution of inclina-

TABLE 3

Structural and paleomagnetic data for site p7

Specimen No.	Bedding Strike/Dip	B component (*Dec / Inc*)
83b	64.0/18.0	183.5/4.6
84b	64.0/18.0	189.9/9.7
85	64.0/18.0	191.3/11.1
86	64.0/18.0	186.0/8.3
87b	64.0/18.0	187.2/14.2
88b	300.0/54.0	180.0/−13.3
90	262.0/26.0	157.3/−30.2
91	271.0/31.0	187.0/−28.0
92	271.0/31.0	191.5/−31.7
93b	289.0/52.0	205.2/−27.3
94b	289.0/52.0	215.5/−9.0
95b	289.0/52.0	206.1/−36.3
96	289.0/52.0	209.0/−40.0
97	105.0/48.0	170.3/−31.3
98	105.0/48.0	178.5/25.2
Means [17]	*in-situ*	189.0/−8.0
		$k = 9$, $\alpha_{95} = 13.7$
	tilt-corrected	188.5/4.0
		$k = 13$, $\alpha_{95} = 11.1$
	60% tilt-corrected	189.0/−0.5
		$k = 22$, $\alpha_{95} = 8.3$
Inclination means [18]	*in-situ*	$I_m = -7.5$
		$k = 6$, $\alpha_{95} = 17.2$
	tilt-corrected	$I_m = 4.0$
		$k = 11$, $\alpha_{95} = 12.2$
	60% tilt-corrected	$I_m = -0.5$
		$k = 34$, $\alpha_{95} = 6.5$

tions only (Fig. 6, bottom) is more spectacular, and the clusterings, reflected by the peaks at 60% untilting, are significant at the 99% confidence level with respect to both uncorrected and tilt-corrected populations. This means that the evolution observed in Fisher statistics mainly depends on inclination evolution, and that the dispersion of declinations, probably due to local rotations about vertical axes, cannot be reduced by the classical tilt correction. Finally, we may note that the mean directions for the uncorrected and tilt-corrected data, as quoted in Table 3, are both consistent with the Permian magnetic field direction at the St Affrique location computed after Van Der Voo [13].

We may draw several conclusions from the above observations.

(1) Seeing as there is no evidence of penetrative deformation of the rocks that could give the magnetization a syn-tectonic-like behaviour in this type of stepwise untilting analysis [e.g. 20,21] it may be assumed that the B magnetic component is acquired during the tilting of the blocks induced by the normal faults.

(2) Based on a comparison with Permian magnetic field direction deduced from the Van der Voo [13] global reference curve the magnetization component determined here is of Permian age. This confirms that the normal fault structure of site p7 developed during the Permian and may indeed reflect a roll-over structure as proposed by Legrand [8].

(3) The "syn-tectonic" behaviour of magnetization with respect to this structure shows that there was a relatively short time delay between bed deposition and magnetization acquisition of the B component, which is thus assumed to be primary. We may therefore assign to it the age of its stratigraphic stage.

4.2 Formation mean directions of magnetization

The results quoted in Table 2 all show southerly directed declinations, and shallow positive or negative inclinations for both the *in-situ* and tilt-corrected data. These are roughly consistent with the Permian field direction in Europe [13]. Following the above analysis of the B component in site p7, and according to similarities in demagnetization behaviour between all the sites and between all the sites and the individual site p7, we assume that the B component is a primary remanent magnetization direction.

Assuming we have isolated the primary magnetization, we have computed mean directions at the chronostratigraphic stage (F1 to F5) level using all specimen data at each stage. The means are given in Table 4, and illustrated in Figs. 7 and 8. Analysis of data from site p7 suggests that the magnetization is acquired during the tilting of the bedding planes. Consequently, it is not possible to state which of the *in-situ* or tilt-corrected directions give the actual paleofield direction. Owing to the generally E–W trend of the beds, and to their weak tilt angle, we may note, however, that there is no significant difference in declination before and after the tilt correction, and therefore this problem of tilt correction has no effect on the study of rotations about a vertical axis.

In Fig. 7, the mean formation directions of the B component are illustrated with the inclination as a function of age (following the time scale of Odin and Odin [22]), and are compared with the

TABLE 4

Formation mean directions of B component and reference magnetic dipole field direction for stable Europe

Formation	n	*In-situ*			Tilt-corrected			Field direction **		Age limits (Ma)
		Dec / Inc	*k*	α_{95}	*Dec / Inc*	*k*	α_{95}	*Dec*	*Inc*	
F1–F2a	29	173.0/5.0	51	4.0	172.5/9.5	28	5.0	194° ± 2°	−1.4° ± 4°	267–281
F2b	8	185.0/−3.0	118	5.1	186.0/−20.5	118	5.1			
F3 *	14	185.0/18.0	29	7.5	183.5/−1.5	38	6.5	195° ± 5°	−10.5° ± 9.7°	246–266
F4/5	21	199.0/−5.0	44	5.0	199.0/−6.5	36	5.5			
F5	18	191.5/−6.5	122	3.0	192.0/−14.5	144	3.0			

* Without site p7. ** Computed at 44°N, 2.75°E after Van Der Voo [13].

inclination of the magnetic dipole field recalculated at the mean position of St Affrique (44°N, 2.75°E) after the reference European apparent polar wander path of Van Der Voo [13] from the Upper Carboniferous to the Lower Jurassic. It is obvious that inclination data are quite scattered in both the *in-situ* and the tilt corrected data: middle/upper Thuringian data are consistent with the reference curve in both data types, lower Thuringian data fit this curve only after tilt correction, and Saxonian and upper Autunian data are closer to the curve before tilt correction, and disperse in the opposite direction on tilt correction. This inconsistency of inclination behaviour is, however, not surprising if we take into account the possible "syn-tectonic" age of magnetization acquisition with respect to the bed tilting that took place during the opening of the basin.

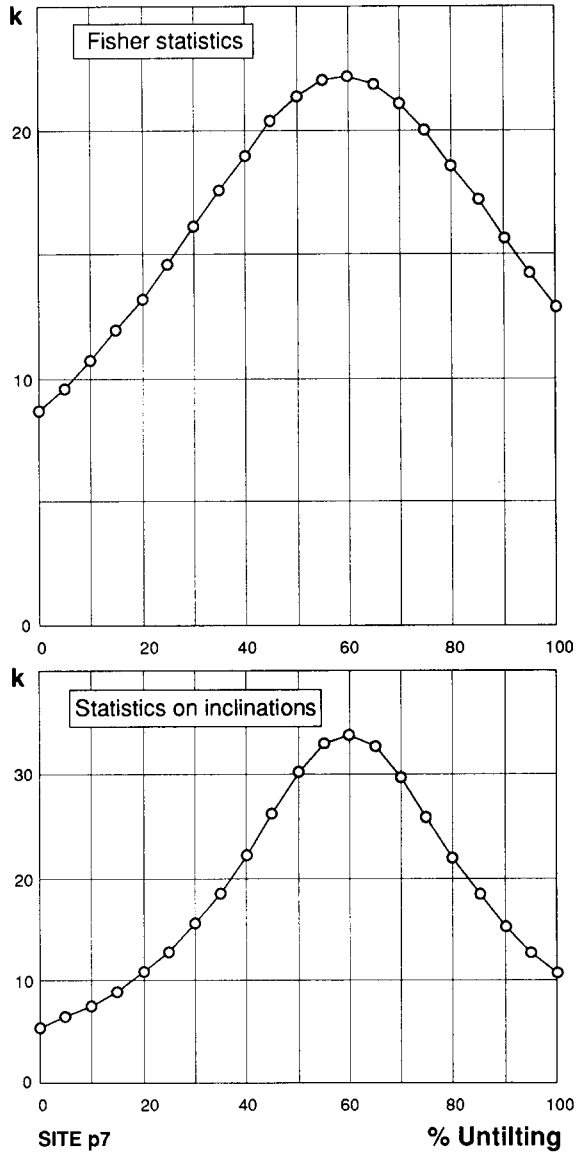


Fig. 6. Stepwise untilting of B components from site p7. *Top* Evolution of Fisher [17] k grouping parameter as a function of untilting percentage. *Bottom* Evolution of the grouping parameter k of the statistics on inclinations of McFadden and Reid [18] as a function of untilting percentage.

The same type of comparison is made for declinations in Fig. 8. As noted above, there is no difference between *in-situ* and tilt-corrected declinations, and in contrast to the inclinations there is a clear trend in the declination data: the older F1–F2a formation (upper Autunian) is rotated counterclockwise by $21° \pm 4.5°$ (computed on the *in-situ* directions); the younger formations F4/5 and F5 (middle/upper Thuringian) are close or superimposed on the reference curve (differences of $4° \pm 6°$ and $3.5° \pm 4.5°$ respectively); the intermediate F2b (Saxonian) and F3 (lower Thuringian) formations show intermediate counterclockwise differences of $10° \pm 7°$ and $10° \pm 9°$, respectively. It thus appears that there is a continuous difference in declinations between our data and the reference curve. This difference averages about 20° for the Lower Permian, 10° for the mid-Permian and 0° for the Upper Permian. On the basis of this consistency, we therefore assume that (1) the whole basin rotated counterclockwise during the Permian, (2) this rotation was progressively recorded through magnetization acquired by the sediments, and (3) the rotation probably ended by the late Permian. Finally, if we suppose a constant rotation rate during the Permian, our data suggest a reasonable rotation speed of about 0.6°/m.y. during this period.

5. Conclusions

We have presented the results of a paleomagnetic study of Permian samples from the St Affrique basin. In order to test for a rotation of this basin about a vertical axis during its formation, as expected from the kinematic model of Van Den Driessche and Brun [2,3] and from our preliminary paleomagnetic study [1], we have sampled twelve sites throughout the stratigraphic se-

quence, spanning over about 35 Ma from the upper Autunian to the upper Thuringian. The results of the paleomagnetic analysis, using thermal demagnetization procedures, may be summarized as follows.

(1) In addition to a low temperature component (A), which is unblocked by 300°C and which conforms to the present-day dipole field direction, a medium temperature (B) component has been isolated and a high temperature (C) component is suspected. On the basis of unblocking temperature spectra and IRM acquisition curves it is believed that C is carried by hematite, while magnetite is the likely carrier of B.

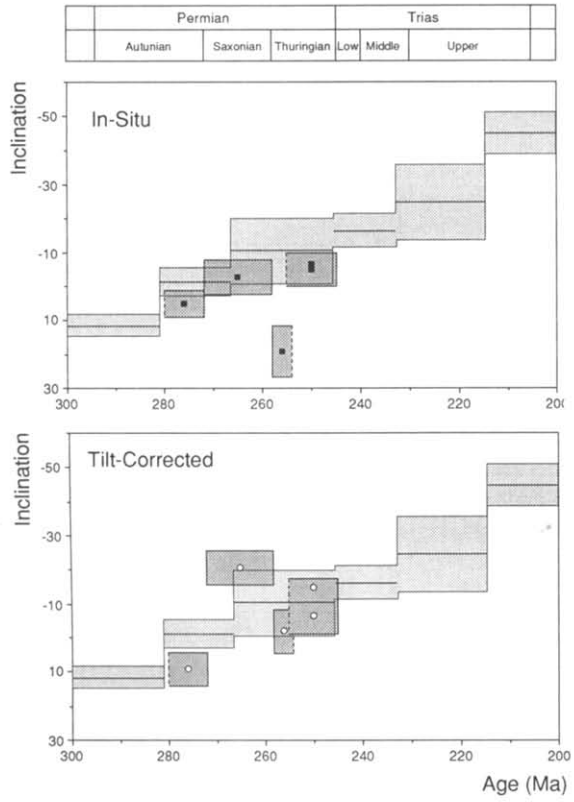


Fig. 7. Formation mean inclination of B component of magnetization as a function of the age (m.y.) of the sampled Permian formations in St Affrique, for *in-situ* (*top*) and tilt-corrected (*bottom*) data. Stratigraphic scale in the upper part of the figure is after Odin and Odin [22]. Our data (dark grey boxes) are here compared to a reference inclination curve (light grey boxes) of the dipolar magnetic field calculated at the St Affrique location, after the reference European apparent polar wander path of Van Der Voo [13].

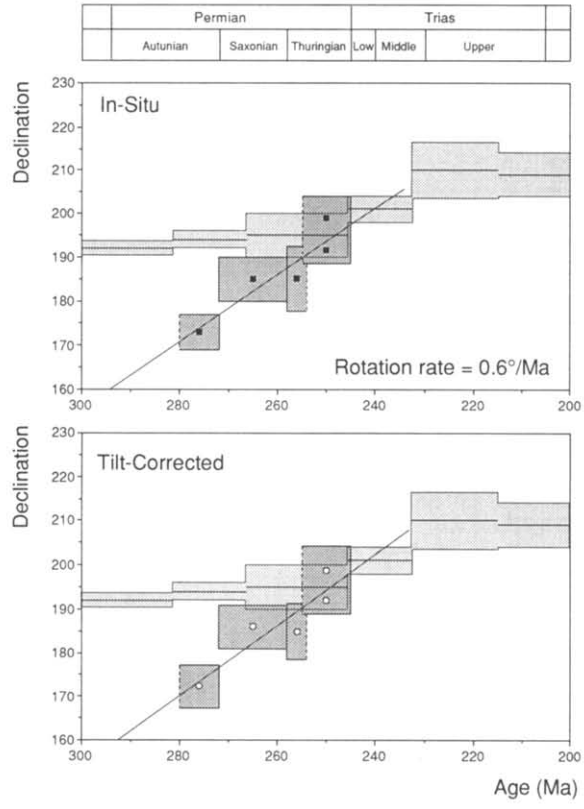


Fig. 8. Same as Fig. 7, but for paleomagnetic declinations of the B component. Least-square best fit lines over experimental points are drawn.

(2) Site p7, in red sandstones of lower Thuringian age, shows a distinctive tilted block geometry which has been interpreted as resulting from the development of a roll-over anticline during basin opening on a listric fault [8]. This syn-sedimentary structure allowed us to test for the age of the B component of magnetization. The vector population significantly clusters by 60% untilting, in a direction consistent with the Permian magnetic dipole field direction. We interpret this as a "syn-tectonic" acquisition age of the B magnetization component, with respect to this structure. B is therefore a primary magnetization acquired early in the sedimentary history.

(3) Because magnetization is probably acquired during the tilting of the beds, and also owing to the slight dipping of bedding planes, we were unable to determine for which data, *in-situ* or tilt-corrected, magnetization gives the paleofield direction. Fortunately, this problem only affects inclinations; declinations are generally

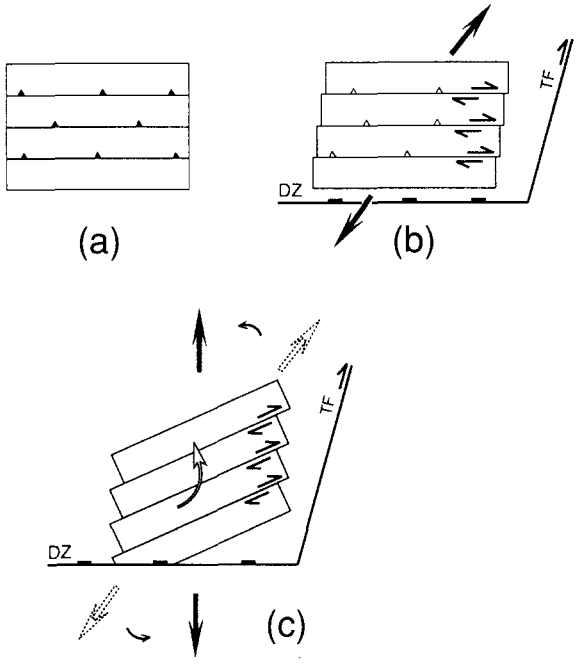


Fig. 9. Kinematic interpretation of the Monts de Lacaune domino-like rotation during Stephano-Permian times. (a) Stage prior to extension. Inferred E–W trending thrusts of the Monts de Lacaune area are indicated. (b) Stephanian. Stage of NE–SW extension (black arrows) with development of the Espinouse detachment zone (*DZ*) and Cévennes transfer fault (*TF*). (c) Permian. Rotation of the direction of extension to a N–S direction, inducing a rotation of the Monts de Lacaune thrusts (curved white arrow), and dextral shear along the thrusts.

identical before and after the tilt correction. We have computed the mean direction of the B component for each stratigraphic stage. If compared to the magnetic field deduced from the stable Europe apparent polar wander path, our data show a large counterclockwise difference in declination for the Lower Permian, no significant difference for the Upper Permian, and an intermediate difference for the mid-Permian.

(4) Because post-Permian deformation played only a minor role in the studied area as a whole, and no significant related structures are observed within the sampling sites, differences recorded in paleomagnetic data cannot be explained by post-Permian tectonism. Instead, the paleomagnetic results are consistent with the idea of a progressive counterclockwise rotation of the St Affrique basin during at least the whole of the Permian. Assuming a constant rotation during this period, we estimate a rotation rate for the basin of about 0.6°/m.y..

Although the mode of deformation within the basement is beyond the scope of this work, we suggest that the reactivation by dextral wrench faulting of some of the Monts de Lacaune thrusts during Stephano-Permian times [23,24] may have accommodated a domino-like rotation of the basement, as shown schematically in Fig. 9. The rotation of both the St Affrique basin and the Monts de Lacaune emphasizes the regional variation from ENE–WSW to N–S of the extension direction throughout Late Carboniferous to Permian in this part of the Hercynian chain [3].

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